

INDEPENDENT EXECUTION + SOURCE AUDIT

KLB v0.6.0

Network Pruning Verification

Does the compact full-SGP4 GPU approach seriously work—and what does it not prove?

CONDITIONALLY VALIDATED

Pruning correctness and GPU feasibility: yes. Shipped acceptance gate and precision sat-nav accuracy: no.

TESTED ON	SCOPE
NVIDIA RTX 5070 Ti Laptop CUDA 12.8 / native sm_120	58 objects × 16 stations 7 days / 10,080 60-second intervals
REPORT DATE	PACKAGE
17 August 2026 Europe/Amsterdam	klb_seedchain_gpu_v0.6.0_network_pruning

Prepared from fresh builds, full CPU/GPU runs, independent counter instrumentation, CUDA sanitizers, Nsight Compute, package-integrity checks, and a real IGS SP3 comparison.

01 / VERDICT

The honest answer: it works, with two important boundaries

The package succeeds at its central engineering idea: reject impossible network relations before doing needless work, while preserving the event set. That is different from saying every shipped acceptance gate passes or that the propagated coordinates are precise enough for GNSS navigation.

2.1187× STATIC RELATION REDUCTION	9,335 EVENT IDENTITIES PRESERVED	0 CUDA SANITIZER ERRORS	56.80 km REAL-SP3 POSITION RMS
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Bottom line: yes for compact SGP4-based network filtering, visibility scheduling and GPU feasibility. No for a clean as-shipped acceptance pass, and no for precision satellite-navigation claims.

Decision matrix

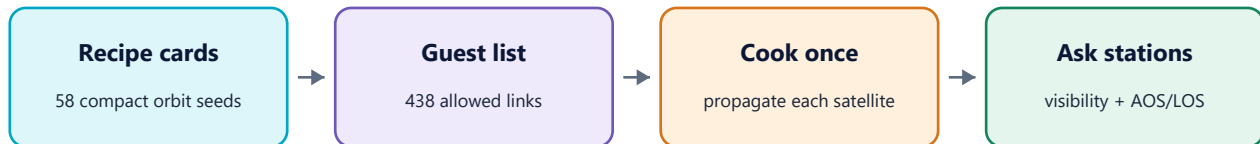
AREA	RESULT	WHAT THE EVIDENCE SAYS
Pruning logic	PASS	All/support/active CPU runs produced the same 9,335 event identities. Independent GPU audit matched all identities and all exact counters.
GPU feasibility	PASS	Full native sm_120 build, laptop-scale 1,048,576-step run, 1.856 GiB dense allocation, no OOM or propagation errors.
GPU safety	PASS	Compute Sanitizer: 0 memory errors, 0 race hazards, 0 synchronization errors.
Release acceptance	FAIL	The executable returns 1 because its 1 ns / 1e-12 CPU-GPU comparator is tighter than normal floating-point CPU/GPU reproducibility.
Precision sat-nav	FAIL	Against a real IGS ultra-rapid SP3 product, RMS position error was 56.80 km; unsuitable for PNT/pseudorange-grade orbit truth.

This is not contradictory. The GPU and CPU can implement the same SGP4 model extremely closely while that model, using the supplied element set and simplified frame treatment, is still far from a precision ephemeris.

02 / ELI5

Think recipe cards, a guest list, and one cook

The compact file does not store millions of satellite coordinates. It stores the ingredients needed to calculate them when asked.



The file stores the recipe, not every future position.

Grouped mode calculates one satellite state, then checks all eligible stations.

What “compression” means here

- The 9,581-byte KSGP file is a compact parametric description: orbit elements, names, metadata, and timeline nodes.
- A seven-day GPU position+velocity table at 60-second spacing occupies 17.84 MiB. The KSGP representation is 1,953× smaller than that table because it delays calculation until query time.
- It is not compression of the original OMM CSV: that source is 8,454 bytes, so KSGP is actually about 13.3% larger. The win is against dense future samples, not against source text.

What pruning means

- Start with every satellite-station invitation: $58 \times 16 = 928$ possible relations.
- Remove invitations that cannot meet the declared range envelope: 711 remain.
- Remove invitations blocked by orbit/service/route policy: 438 remain.
- The crucial proof is that the remaining list still produces the same 9,335 AOS/LOS events as checking every invitation.

Grouped GPU mode is the sensible design: calculate one satellite position, then ask every eligible station. Pair-expanded mode recalculates the same satellite for multiple stations and is mainly a comparison baseline.

03 / METHOD

What was actually tested

No verdict here is based only on a successful process launch. The audit separated package integrity, model correctness, GPU equivalence, performance, robustness, coverage, and physical accuracy.

LAYER	TEST	EVIDENCE
Package	Filesystem + ZIP SHA-256 manifest	251/251 entries matched in both locations
Toolchain	Fresh CMake/MSVC/CUDA configure and Release build	CUDA 12.8, CC 12.0, sm_120 cubin + compute_120 PTX
Unit/regression	CTest Release	4/4 executables passed
CPU oracle	Full 7-day all/support/active modes	9.354M / 7.167M / 4.415M relation intervals
GPU paths	pair_all, pair_support, pair_active, grouped_all, grouped_active, dense	Independent exact event/counter capture
Robustness	memcheck, racecheck, synccheck, truncation, malformed stations	Expected clean/rejected outcomes
Performance	11-repeat file run + 7-repeat laptop run	p50/p95/p99 CSV plus telemetry
Kernel	Nsight Compute full set + cuobjdump	occupancy, registers, cache, branch and ISA data
Physical truth	Real IGS ultra-rapid SP3, 32 GPS, 192 epochs	6,144 coordinate samples + 49,152 station endpoints

The acceptance logic I applied

- A correctness pass requires identical event identities and exact intended counters after pruning—not merely similar totals.
- A GPU feasibility pass requires a native architecture build, sustained execution, safe memory behavior, and practical resource use.
- A physical-accuracy pass requires comparison to external orbit truth. CPU/GPU agreement alone cannot establish it.
- A claim-coverage pass requires the data to exercise the claimed orbit classes, station profiles, policies, routes, and event-producing cases.

Temporary diagnostic instrumentation was used only to expose hidden counters and numeric deltas. It was removed, the original source SHA-256 was restored exactly, and the original executable was rebuilt.

04 / PACKAGE + BUILD

The laptop hardware is adequate; automation discovery is the weak link

The code built and ran natively on the RTX 5070 Ti Laptop GPU. The failures seen in the automatic path were Windows/CMake integration issues, not lack of GPU capability and not a flaw in the pruning concept.

251/251 MANIFEST FILES MATCH	4/4 CTEST PASS	sm_120 NATIVE CUBIN	12,227 MiB GPU VRAM
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AREA	RESULT	WHAT THE EVIDENCE SAYS
Filesystem manifest	PASS	251 present, 0 missing, 0 mismatched, 0 malformed.
ZIP manifest	PASS	1,901,903-byte archive; all 251 declared entries hash correctly inside the archive.
Long build path	CONDITIONAL	MSBuild FileTracker failed at the original deep path. This is a Windows path/tooling limitation.
Short auto-configure	FAIL	CMake still did not auto-discover CUDA 12.8, but the script propagated the failure correctly.
Explicit CUDA configure	PASS	Using -T cuda=... and CMAKE_CUDA_COMPILER produced a clean Release build.
Binary architecture	PASS	cuobjdump found native sm_120 code plus compute_120 PTX.

Compiled kernel resources

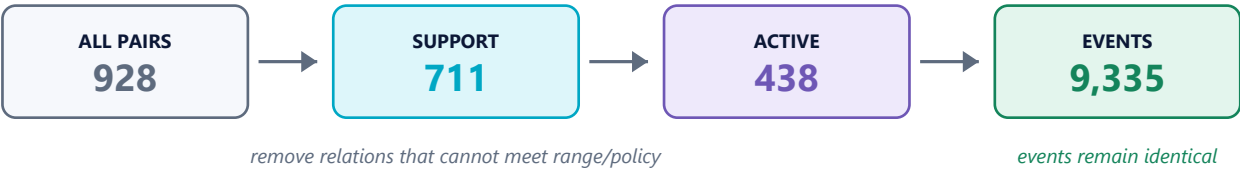
KERNEL	REGISTERS	STACK	SPILLS
pair_query_kernel	152	48 B	0
grouped_query_kernel	154	48 B	0
materialize_kernel	142	48 B	0
dense_query_kernel	74	0 B	0

Answer to the earlier hardware question: the laptop was not what was lacking. The explicit CUDA build proves the hardware and compiler path work; automatic toolkit discovery and path length are the build-automation defects.

05 / PRUNING CORRECTNESS

The core pruning proof passed exactly

Every relation mode was evaluated over the complete seven-day, 60-second challenge. Static pruning removed work without removing any valid AOS/LOS event.



9,354,240	7,166,880	4,415,040	2.1187×
ALL RELATION INTERVALS	SUPPORT INTERVALS	ACTIVE INTERVALS	TOTAL PRUNING GAIN

EXACT COUNTER	ALL	SUPPORT	ACTIVE / GROUPED
relation intervals	9,354,240	7,166,880	4,415,040
propagated intervals	9,354,240	7,166,880	4,415,040 pair / 584,640 grouped
supported	5,932,861	5,932,861	3,970,997 active
compatible + boundary	3,970,997	3,970,997	3,970,997
visible	1,101,511	1,101,511	1,101,511
AOS / LOS / total events	4,653 / 4,682 / 9,335	same	same
propagation errors	0	0	0

Independent audit discoveries

- The release's displayed "counter match" helper omits relation intervals, propagated intervals, and supported intervals. My instrumentation checked them explicitly; they match the intended values.
- The default benchmark times pair_support but does not compare its event vector. My independent capture showed pair_support and pair_active events match exactly.
- Grouped active reduces orbit propagations by 7.5517× relative to pair-active, because each object/time state is reused across stations.

06 / CPU-GPU NUMERICS

Why the executable says FAIL even though the results verify

The release comparator requires crossing times within 1 nanosecond and guards within 1e-12. CPU and GPU execute floating-point transcendental operations differently, so these tolerances are not realistic for cross-architecture reproducibility.

9,335/9,335 EVENT IDENTITIES MATCH	29.23 μs CROSSING-TIME RMS DELTA	343.26 μs WORST TIME DELTA	9.56e-8 WORST GUARD DELTA
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The GPU run is not a false pass; it is a false failure at the comparator boundary. Counts and identities agree, while every event exceeds an unnecessarily strict tolerance by a very small numerical amount.

Observed delta distribution

METRIC	P50	P95	P99	MAX
crossing time	16.98 μs	54.79 μs	78.67 μs	343.26 μs
visibility guard	3.01e-9	4.24e-8	6.11e-8	9.56e-8

What must change

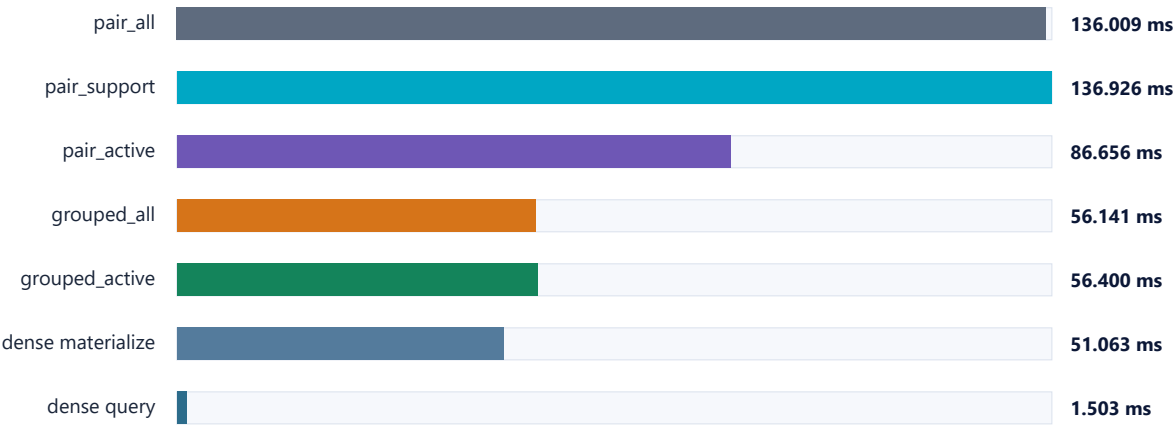
- Define a documented engineering tolerance based on the scheduling requirement—likely milliseconds for crossing interpolation, not nanoseconds.
- Keep exact identity/count checks, exact integer counter checks, no-truncation checks, and zero-propagation-error checks.
- Report maximum and percentile numeric deltas instead of collapsing all agreement into one boolean.
- Add pair_support events and the omitted counters to the normal release acceptance output so the independent diagnostic patch is unnecessary.

Until that is done, the honest release status is: algorithmically validated, acceptance gate failing as shipped.

07 / SEVEN-DAY PERFORMANCE

Grouped reuse beats repeated pair work

These are medians from the full 10,080-interval, seven-day file preset. Lower is better.



1.57×	1.54×	37.52×	52.57 ms
PAIR ACTIVE VS ALL	GROUPED VS ACTIVE PAIR	RESIDENT DENSE QUERY	DENSE END-TO-END

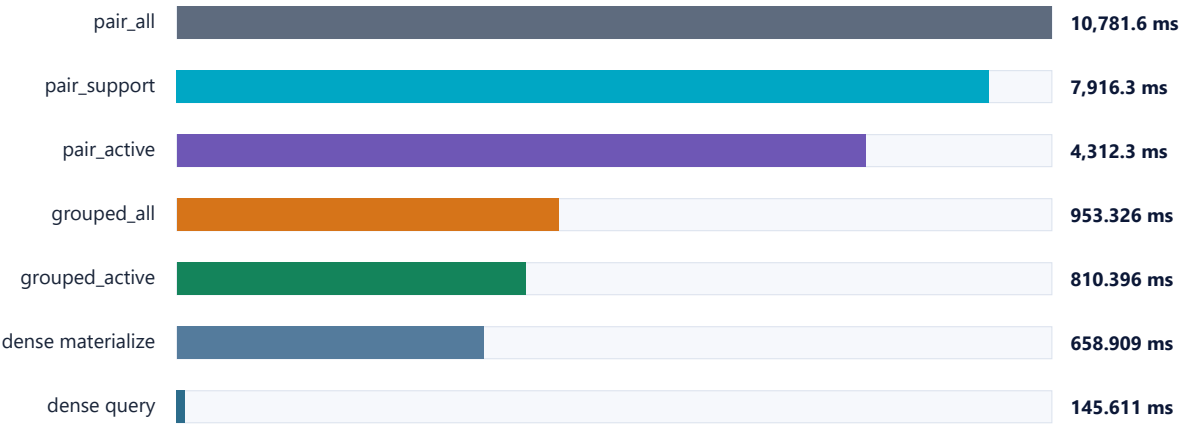
- pair_active: 86.66 ms, down from pair_all at 136.01 ms. Static active pruning produces a real 1.57× speedup in the pair-expanded design.
- pair_support: 136.93 ms, slightly slower than pair_all. Removing only 23% of pairs does not offset launch/divergence/occupancy effects at this small horizon.
- grouped_active: 56.40 ms. It is 1.54× faster than pair_active because it propagates each object/time once, then reuses the state.
- grouped_all and grouped_active are almost identical at this scale. Orbit propagation and the small eight-block grid dominate, so pruning station checks alone is not the primary win.
- materialize + dense query is 52.57 ms, about 6.8% faster than grouped_active even for one query. Dense is the latency winner when the fixed horizon comfortably fits memory.

The compact method's strongest case is low storage, flexible horizons, and avoiding persistent dense state—not absolute single-query latency when a dense table already fits.

08 / LAPTOP-SCALE STRESS

A million time steps ran stably at sustained load

The prescribed laptop preset used 1,048,576 one-second intervals. The dense table alone was 1.856 GiB. Seven timed repetitions were completed for every strategy.



99.61% AVG GPU UTILIZATION	74 °C PEAK TEMPERATURE	85.06 W PEAK POWER	2,472 MiB PEAK USED VRAM
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MODE	P50	P95	KEY INTERPRETATION
pair_all	10,781.62 ms	10,783.31 ms	baseline repeated propagation
pair_active	4,312.30 ms	4,312.87 ms	2.50× faster than pair_all
grouped_active	810.40 ms	811.64 ms	5.32× faster than pair_active
dense end-to-end	804.52 ms	≈805.54 ms	0.7% faster than grouped_active
resident dense query	145.61 ms	145.62 ms	5.57× faster after materialization

Stress verdict

- No out-of-memory event, GPU reset, propagation error, or sanitizer fault occurred.
- 777 telemetry samples covered the run. Average power was 73.70 W; average temperature was 70.34 °C; SM clock averaged 2,757 MHz.
- Dense event equivalence was intentionally skipped by the release on this preset because validation was capped at 10,080 intervals. Full dense equivalence was already proven on the seven-day file preset.

09 / STORAGE + FEASIBILITY

The representation is extremely compact versus sampled trajectories

The package's compression story is real only when stated against the correct baseline: future sampled states.

9,581 B KSGP CONTAINER	17.84 MiB 7-DAY DENSE GPU TABLE	1,953× 7-DAY DENSE / KSGP	203,127× LAPTOP DENSE / KSGP
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REPRESENTATION	SIZE	WHAT IT CONTAINS	TRADE-OFF
source OMM CSV	8,454 B	human-readable mean elements	smallest source; parsing/initialization required
KSGP1	9,581 B	58 fixed seeds + names + compiled/timeline metadata	GPU-ready; 13.3% larger than source CSV
dense float4	8.92 MiB	position-only 16-byte states	very fast reuse; no velocity
dense position+velocity	17.84 MiB	32-byte state per object/time	fastest query; fixed horizon and memory
laptop dense position+velocity	1.856 GiB	1,048,576 samples	still feasible here, but scales linearly

Feasibility boundaries

- The compact approach wins decisively when you need many possible horizons, distribute small orbit datasets, cannot reserve dense memory, or queries are sparse/streaming.
- Dense materialization wins latency when the same fixed horizon is queried repeatedly and memory is available. In both tested horizons it already broke even within the first query.
- Memory grows as objects × samples × bytes-per-state. KSGP size is nearly independent of sample count; computation grows with the samples actually evaluated.
- KSGP is a model-backed representation, not lossless compression of an exact future trajectory. Its physical fidelity is bounded by the SGP4 inputs and frame/time treatment.

Serious answer on compression: yes, the deferred-state representation is real and useful. But advertising “1,953× compression” must name the dense sampled-state baseline and must not imply higher physical accuracy.

10 / EXACT COVERAGE

The test is broad, but not literally universal

The pair plan covers the full 58×16 Cartesian product and all declared orbit classes and route sectors. Event-producing coverage is narrower, especially for GEO relay cases.

928/928 UNIQUE PAIRS PRESENT	54/58 OBJECTS WITH EVENTS	14/16 STATIONS WITH EVENTS	369/438 ACTIVE PAIRS WITH EVENTS
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ORBIT	OBJECTS	ACTIVE	ACTIVE + EVENTS	EVENTS	OBJECTS + EVENTS
LEO	11	99	74	3,625	11
MEO	32	256	256	5,191	32
GEO	8	48	4	56	4
HEO	7	35	35	463	7

What is covered

- LEO 11, MEO 32, GEO 8, HEO 7; service classes include NAV, RELAY, SCIENCE, EARTH_OBS, and CREWED combinations.
- All route sectors 0–5 appear in the event set; event interval indices span 3 through 10,077.
- All 9,335 events map only to active pairs. No pruned pair reappears in the oracle output.

Gaps that matter

- Four TDRS GEO objects generated no events, and only 4 of 48 active GEO pairs generated events. This is not deep GEO event diversity.
- Singapore Relay and Equatorial Relay profiles generated no events. Sixty-nine active pairs generated none during the seven-day window.
- The network fixture contains no half-day resonant objects. Separate core SGP4/Vallado tests cover the half-day branch, but not through this network-pruning workload.
- The data is a deterministic challenge fixture, not a probabilistic survey of all real station networks, masks, ranges, outages, or policy combinations.

11 / PHYSICAL ACCURACY

Full SGP4 runs—but it is not precision sat-nav truth

A real external IGS ultra-rapid SP3 file was compared against the package's GPS predictions: 32 satellites, 192 fifteen-minute epochs, 6,144 position samples over 48 hours.

56.80 km	54.77 km	60.83 km	126.67 km
POSITION RMS ERROR	MEDIAN ERROR	P95 ERROR	MAX ERROR

This result rules out precision GNSS/PNT use. Receiver navigation and pseudorange work require precise time/frames and metre-to-centimetre-class orbit products—not tens of kilometres.

Visibility impact over the same SP3 window

CHECK	SGP4	SP3	AGREEMENT / DELTA
station endpoints	14,556 visible	14,550 visible	49,124/49,152 agree (99.943%)
support predicate	—	—	2 endpoint disagreements
coarse 15-min crossings	645	650	588 common identities
common crossing time	—	—	18.80 s RMS; 49.65 s max
unmatched crossing identities	57 only SGP4	62 only SP3	boundary/interval sensitivity

How both facts can be true

- A 50–60 km difference is small compared with a roughly 26,600 km GPS orbital radius, so most above/below-mask endpoints remain unchanged; near a boundary, the same error shifts crossing time or interval identity.
- The official IGS ultra-rapid product contains 24 hours observed plus 24 hours predicted. RMS was 56.25 km in the observed half and 57.34 km in the predicted half, so prediction alone does not explain the result.
- TLE/OMM SGP4 is useful for coarse scheduling and catalog tracking. It is not the orbit/time/frame stack used for precision satellite navigation.

Reference: [IGS Products](#); orbit file IGS00PSULT_20262271800_02D_15M_ORB.SP3 (retrieved from the UCSD IGS archive mirror).

12 / GPU ENGINEERING

The kernel is compute-heavy, register-heavy, and under-filled on the smoke grid

Nsight Compute profiled grouped_active after skipping the grouped_all launches. The smoke case uses only eight blocks, so these numbers diagnose shape and resource pressure rather than laptop-scale throughput.

154 REGISTERS / THREAD	25% THEORETICAL OCCUPANCY	4.58% ACHIEVED OCCUPANCY	97.72% BRANCH EFFICIENCY
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METRIC	VALUE	INTERPRETATION
grid / block	8 × 128	only 0.06 waves/SM; too small to fill GPU
register limit	3 blocks/SM	154 registers constrain theoretical occupancy
SM busy	44.89% active cycles	FP64 is the most-used pipeline
memory throughput	0.12% peak	not DRAM-bandwidth-bound in this profile
L1/TEX hit rate	99.86%	hot working set; memory locality is good
eligible warps/scheduler	0.08	latency hiding is weak in tiny grid
stack / spills	48 B / 0	stack exists, but no register spills reported

Engineering implications

- Grouped reuse is the right algorithmic direction. It removes repeated propagation before micro-optimizing station predicates.
- For small object counts, launch shape and available parallelism dominate. Consider batching independent networks, windows, or epochs to expose more blocks.
- Reducing register pressure may raise occupancy, but SGP4's double-precision state is intrinsically register-intensive. Validate accuracy before trading doubles for floats.
- Pruning yields larger timing benefits on the million-step workload because there is enough work to amortize overhead and expose its reduced relation count.

Do not interpret the 4.58% smoke occupancy as a failure of the laptop. The sustained run reached 99.6% GPU utilization; the profile shows that this tiny grid under-fills the device, while the long grid does not.

13 / INTENDED-ASPECT CHECKLIST

What passed, what is conditional, and what remains unproven

This matrix maps the package's intended challenge areas to fresh evidence rather than treating a single console summary as the entire test.

AREA	RESULT	WHAT THE EVIDENCE SAYS
All four orbit classes	PASS	LEO/MEO/GEO/HEO all present; core half-day resonance is separate, not exercised by network data.
Station policy pruning	PASS	16 typed profiles; exact active-mask invariant; all events belong only to active pairs.
Support pruning safety	PASS	All/support event identities and survivor counts match; no lost valid event.
CPU active pruning	PASS	All/active produce identical 9,335-event identity set and exact expected counters.
GPU pair equivalence	PASS	All, support and active identities/counters independently captured and exact.
Grouped equivalence	PASS	Pair/grouped event identities and all audited counters match.
Dense equivalence	PASS	Full seven-day grouped/dense validation passes; long preset deliberately validates only prefix.
No errors/truncation	PASS	Zero propagation errors; capacity=1 rejects explicitly as truncated.
Input rejection	PASS	Duplicate station IDs and invalid latitude are rejected with specific errors.
CUDA safety	PASS	memcheck/racecheck/synccheck all report zero findings.
Windows build automation	CONDITIONAL	Correct failure propagation, but CUDA auto-discovery and deep path remain unreliable.
CPU/GPU release gate	FAIL	All identities match; release tolerance is nevertheless too strict and returns failure.
Physical orbit accuracy	FAIL	Real-SP3 trajectory error is tens of kilometres; not precision navigation.
Universal workload claim	CONDITIONAL	Broad deterministic fixture, but eventless GEO/station cases and no network half-day resonance.

Overall engineering classification: prototype validated for its pruning and compact-compute purpose; release acceptance and precision-navigation positioning require corrective work.

14 / NEXT ACTIONS + EVIDENCE

How to turn this into a clean release pass

The central algorithm does not need to be discarded. The next work is to make acceptance honest, expand coverage, and separate coarse scheduling claims from precision navigation claims.

PRIORITY	ACTION	ACCEPTANCE TARGET
P0	Replace 1 ns / 1e-12 cross-device gate with documented engineering tolerances; retain exact identity and integer checks.	Executable exits 0 on current RTX while reporting p50/p95/p99/max deltas.
P0	Add relation, propagation, supported counters and pair_support event comparison to normal validation.	No diagnostic patch required to prove all intended counters.
P1	Fix CUDA 12.8 auto-discovery and shorten/normalize Windows build paths.	Official script configures and builds from the distributed path.
P1	Keep real SP3 regression separate from implementation equivalence; add EOP/time/frame documentation.	Claims explicitly say coarse SGP4 scheduling, not precision PNT.
P1	Add event-producing GEO/relay and half-day resonant network cases.	Every intended branch produces observable network events.
P2	Batch small workloads or tune register pressure after correctness is locked.	Higher achieved occupancy without changing event/crossing accuracy.

Key evidence produced by this run

ARTIFACT	PURPOSE
verification_v060_cpu_verify_console.txt	full seven-day CPU oracle and pruning equivalence
verification_v060_exact_counter_coverage_console.txt	independent all/support/active/grouped/dense counters
verification_v060_gpu_cpu_mismatch_diagnostic_console.txt	identity and numeric-delta diagnosis
verification_v060_gpu_file_results.csv	11-repeat full-horizon GPU metrics
verification_v060_gpu_laptop_results.csv	7-repeat million-step GPU metrics
verification_v060_gpu_laptop_telemetry.csv	777 temperature/power/clock/utilization samples
verification_v060_ncu_grouped_active.ncu-rep	Nsight Compute full kernel profile
verification_v060_coverage_summary.json	pair/event/orbit/station/SP3 coverage audit
verification_v060_sp3_comparison.csv	6,144 real-SP3 coordinate deltas
verification_v060_manifest_console.txt	251-entry filesystem integrity result

Final answer

Yes, it seriously works for its actual purpose: compact, on-demand full-SGP4 propagation plus static network pruning and grouped GPU visibility evaluation. It is not yet a clean release pass because the comparator rejects harmless CPU/GPU drift, and it is not a precision satellite-navigation solution because external orbit error is far too large.

All metrics in this report were generated locally from the named package and hardware on 17 August 2026. The original declared package files and ZIP passed SHA-256 verification; newly created verification artifacts are intentionally outside the supplied manifest.